

Physics Study Notes

Chapter 1: Newton's Laws of Motion

Newton's First Law: An object remains at rest or in uniform motion unless acted upon by an external force.

Newton's Second Law: Force equals mass times acceleration ($F = ma$).

Newton's Third Law: For every action, there is an equal and opposite reaction.

Chapter 2: Energy

Kinetic Energy = $\frac{1}{2} mv^2$

Potential Energy = mgh

Chapter 3: Momentum

Momentum = mass $\times$ velocity.

Law of Conservation of Momentum: Total momentum remains constant in an isolated system.